

Daniel Jiarui Zou

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EDUCATION

University of Michigan

PhD in Statistics (Advised by Moulinath Banerjee and Yuekai Sun; GPA=3.98/4.00)

Ann Arbor, Michigan

Sept. 2023 – Present

University of Michigan

MA in Statistics

Ann Arbor, Michigan

Sept. 2023 – May 2025

Reed College

BA in Mathematics - Computer Science (GPA = 3.85/4.00, Phi Beta Kappa)

Portland, Oregon

Sept. 2019 – May 2023

RESEARCH

Performative Prediction in Score-Explained Heterogenous Treatment Effect Models

- Collaborating with Moulinath Banerjee and Ya'acov Ritov (May 2024 - Present)
- Investigating the setting of score-explained heterogenous treatment effect models, specifically those where the treatment assignment is determined by a threshold on some score. Recent work in this area focuses on estimating average treatment effects. Instead, we are interested in optimizing the score threshold to maximize utility for the organization. We propose an algorithm for this optimization and establish asymptotic normality results for our estimator.
- Consider a performative setting, where setting a score threshold changes the distribution of the agents' covariates and scores. (Ex. Pre-med students adjusting their preparation to adapt to different MCAT score thresholds.) Under some conditions on this distribution shift, we establish performative stability results.

Estimation and Sensitivity Analysis under Distribution Shift

- Collaborating with Yuekai Sun and Subha Maity (May 2024 - Present)
- Studying the problem of missing not at random (MNAR) datasets with discrete outcomes with exponential tilt models. We establish sufficient conditions for identifiability of tilt parameters and propose an algorithm to estimate them. We then propose importance weighted, imputation based, and doubly robust estimators for any mean functions of interest. In an experiment with the Waterbirds dataset, we utilize our tilt framework to perform unsupervised transfer learning, and achieve a prediction performance that is comparable to a gold standard.
- Investigates a distribution shift setting where the target dataset is unlabeled. Most methods for such out-of-distribution (OOD) accuracy estimation assume covariate shift. We propose a method to evaluate these OOD accuracy estimators in the target domain, without any labeled observations, by comparing their sensitivity to certain perturbations away from the covariate shift assumption.

Past Research Projects

- Differentially Private Weighted Linear Regression. Senior Thesis at Reed College. Advised by Leonard Wainstein and Ira Globus-Harris. 2022-2023.
- Differentially Private Statistics REU. Advised by Adam Groce and Andrew Bray at Reed College. Summer 2022.
- Markov Chains REU. Advised by Iddo Ben-Ari at the University of Connecticut. Summer 2021.

PRESENTATIONS

Invited Talks

- “Distribution Matching for Transfer Learning under Conditional Label Shift” at the 2025 Michigan Student Symposium for Interdisciplinary Statistical Sciences; Best Poster Award. March 2025.
- “Finetuning Transformer-based Symbolic Regression for Domain Adaptation” at the LLMs and Transformers Course Poster Session. Dec. 2024
- “Differentially Private Weighted Linear Regression” at the Reed College Mathematics Colloquium. April 2023
- “Differentially Private Data Analysis via Edit Distance” at Reed College Mathematics and Computer Science Colloquiums. Oct. 2022.
- “Differentially Private Data Analysis via Edit Distance” at the Reed College Summer Research Poster Session. Oct. 2022.
- “Coalescing Random Walks on an n-cycle” at the Reed College Mathematics Colloquium. September 2021.
- “Coalescing Random Walks on an n-cycle” at the 2021 REU Virtual Conference. August 2021.

AWARDS

Academic

- Best Poster Award. The Michigan Student Symposium for Interdisciplinary Statistical Sciences (2025).
- Outstanding First Year Student, University of Michigan Department of Statistics PhD Program
- Rackham First Year Fellowship (Fall 2023)
- Commendation for Excellence in Scholarship 2019-2020, 2021-2022, 2022-2023, Reed College

SERVICE

Committees

- Department Culture Committee (Winter 2025/ Fall 2025): Participated in recurrent meetings to discuss topics relating to department culture, including diversity, equity, and inclusion, with leaders of the Department of Statistics at the University of Michigan.
- Seminar Committee (Fall 2025): Participated in meetings with faculty members in the Department of Statistics at the University of Michigan to propose and invite candidate speakers for the department seminar.

Events

- Internship Panel (Fall 2025): Organized and hosted the University of Michigan's Statistics Department's annual internship panel
- Statistics Student Seminar (Fall 2025): Responsible for organizing weekly seminars targeted to graduate students of the Department of Statistics at the University of Michigan.
- StatsPhD Panel (Fall 2024): Spoke at a panel for undergraduates around the US regarding graduate study in Statistics

TEACHING

University of Michigan

- Introduction to Probability and Statistics (STATS 412, Undergraduate Level - Fall 2025)
- Regression and Data Analysis (STATS 513, Graduate Level - Winter 2025)
- Statistics Capstone Seminar (STATS 485, Undergraduate Level, Writing Course - Fall 2024)
- Honors Introduction to Statistics (STATS 280, Undergraduate Level - Winter 2024)
- Introduction to Statistics (STATS 250, Undergraduate Level - Fall 2023)

Reed College

- Computability and Complexity (CSCI 387, Undergraduate Level)
- Algorithms and Data Structures (CSCI 382, Undergraduate Level)
- Linear Algebra (MATH 201, Undergraduate Level)
- Probability (MATH 391, Undergraduate Level)
- Introduction to Probability and Statistics (MATH 141, Undergraduate Level)

SKILLS AND LANGUAGES

Skills

- Computing: Python, R/RStudio (base and tidyverse), Excel
- Publishing: $\text{\LaTeX}$, knitr
- Other: Git/Github

Languages

- English: Native Speaker
- Chinese: Native Speaker